Math for CS & AI: Homework 5

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Problem 3.3

Prove Theorem 3.17 and Theorem 3.18:

$$\operatorname{Var}(cX) = c^2 \operatorname{Var}(X), \tag{3.17}$$
$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y). \tag{3.18}$$

Answer:

For Theorem 3.17, we have:

$$\begin{aligned}
\operatorname{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\
&= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\
&= c^2\left(\mathbb{E}[X^2] - \mathbb{E}[X]^2\right) \\
&= c^2 \operatorname{Var}(X).
\end{aligned}$$

For Theorem 3.18, we have:

$$\begin{aligned}
\operatorname{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - \mathbb{E}[X+Y]^2 \\
&= \left(\mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2]\right) - \left(\mathbb{E}[X]^2 + \mathbb{E}[Y]^2 + 2\mathbb{E}[X]\mathbb{E}[Y]\right) \\
&= \left(\mathbb{E}[X^2] - \mathbb{E}[X]^2\right) + \left(\mathbb{E}[Y^2] - \mathbb{E}[Y]^2\right) \\
&= \operatorname{Var}(X) + \operatorname{Var}(Y),
\end{aligned}$$

where we used the property that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ for independent random variables X and Y.

Problem 3.4

Let $X_1, \ldots, X_n$ be i.i.d. zero-mean random variables, i.e., $\mathbb{E}[X_i] = 0$ for all $i \in [n]$. Let $\mu_m := \mathbb{E}[X_i^m]$ denote the m-th raw moment of an arbitrary X_i. Let X be the sample average of $X_1, \ldots, X_n$. Express $\mathbb{E}[X^3]$, $\mathbb{E}[X^4]$, and $\mathbb{E}[X^5]$ in terms of μ_2, μ_3, μ_4, μ_5.

Answer:

For $\mathbb{E}[X^3]$, we proceed as follows:

$$\begin{aligned}
\mathbb{E}[X^3] &= \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^3\right] \\
&= \frac{1}{n^3}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n}\mathbb{E}[X_i X_j X_k] && \text{(by the linearity of expectation)} \\
&= \frac{1}{n^3}\left(n\mu_3 + 3n(n-1)\mu_2\mu_1 + n(n-1)(n-2)\mu_1^3\right) && \text{(separate the independent variables)} \\
&= \frac{1}{n^2}\mu_3. && \text{(substituting } \mu_1 = 0\text{)}
\end{aligned}$$

Similarly, we derive the expressions for $\mathbb{E}[X^4]$ and $\mathbb{E}[X^5]$ by applying the linearity of expectation, grouping by index equality, using independence of distinct X_i, and the given condition $\mu_1 = 0$:

$$\mathbb{E}[X^4] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^4\right] = \frac{1}{n^3}\left(\mu_4 + 3(n-1)\mu_2^2\right),$$

$$\mathbb{E}[X^5] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^5\right] = \frac{1}{n^4}\left(\mu_5 + 10(n-1)\mu_2\mu_3\right).$$

Problem 3.5

Define the L^p-norm of a random variable X by

$$\|X\|_p := \mathbb{E}\left[|X|^p\right]^{1/p}.$$

Use Jensen's inequality to prove the following inequalities for any real numbers $p, q > 0$ and any pair of random variables X, Y defined on the same finite discrete probability space:

(a) Lyapunov's inequality (L^p-norm increases with p):

$$\|X\|_p \leq \|X\|_q \quad \text{if } p \leq q.$$

(b) Hölder's inequality ("product $\leq$ product"):

$$\|XY\|_1 \leq \|X\|_p \|Y\|_q \quad \text{if } 1/p + 1/q = 1.$$

(c) Young's inequality ("product $\leq$ sum"):

$$\|XY\|_1 \leq \frac{1}{p}\|X\|_p^p + \frac{1}{q}\|Y\|_q^q \quad \text{if } 1/p + 1/q = 1.$$

(d) Minkowski's inequality ("sum $\leq$ sum"):

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p \quad \text{if } p \geq 1.$$

Answer:

Since the norms are defined using absolute values, we may assume without loss of generality that X and Y are non-negative random variables, and are not always zero, to avoid trivial edge cases.

For Lyapunov's inequality (a), let $f(x) = x^{q/p}$. Since $q/p \geq 1$, $f(x)$ is convex on $[0, +\infty)$. Therefore, for the random variable X^p, we apply Jensen's inequality:

$$\begin{aligned}
\mathbb{E}[f(X^p)] &\geq f(\mathbb{E}[X^p]), \\
\mathbb{E}[X^q] &\geq \mathbb{E}[X^p]^{q/p}, \\
\mathbb{E}[X^q]^{1/q} &\geq \mathbb{E}[X^p]^{1/p}, \\
\|X\|_q &\geq \|X\|_p.
\end{aligned}$$

For Hölder's inequality (b), note that $1/p + 1/q = 1$ implies $p, q \geq 1$. Let X and Y be random variables on the probability space (Ω, p_0), we define a new probability mass function $p_1 : \Omega \to [0, 1]$ by:

$$p_1(\omega) = p_0(\omega) \cdot \frac{Y^q(\omega)}{\mathbb{E}[Y^q]}.$$

It follows directly from the definition of expectation that $\sum_{\omega \in \Omega} p_1(\omega) = 1$.

Problem 3.5 continued on next page...

We expand $\mathbb{E}[XY]$ by definition as follows:

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{\omega\in\Omega} X(\omega)Y(\omega)p_0(\omega) \\ &= \sum_{\omega\in\Omega} X(\omega)\left(\frac{p_1(\omega)}{p_0(\omega)}\cdot\mathbb{E}[Y^q]\right)^{1/q}\cdot\left(\frac{p_0(\omega)}{p_1(\omega)}\cdot p_1(\omega)\right) \\ &= \mathbb{E}[Y^q]^{1/q}\sum_{\omega\in\Omega} X(\omega)\left(\frac{p_1(\omega)}{p_0(\omega)}\right)^{-1/p}\cdot p_1(\omega).\end{aligned}$$

Here, we define a new random variable $X'(\omega) := X(\omega)\cdot\left(\frac{p_1(\omega)}{p_0(\omega)}\right)^{-1/p}$, and let $\mathbb{E}'[X']$ denote the expectation of X' on the probability space (Ω, p_1). Thus,

$$\mathbb{E}[XY] = \mathbb{E}[Y^q]^{1/q}\cdot\mathbb{E}'[X'].$$

For the convex function $f(x) = x^p$, we apply Jensen's inequality to obtain:

$$\begin{aligned}\mathbb{E}[XY]^p &= \mathbb{E}[Y^q]^{p/q}\cdot\mathbb{E}'[X']^p \\ &\le \mathbb{E}[Y^q]^{p/q}\cdot\mathbb{E}'[X'^p] \\ &= \mathbb{E}[Y^q]^{p/q}\sum_{\omega\in\Omega} X^p(\omega)\left(\frac{p_1(\omega)}{p_0(\omega)}\right)^{-1}\cdot p_1(\omega) \\ &= \mathbb{E}[Y^q]^{p/q}\sum_{\omega\in\Omega} X^p(\omega)p_0(\omega) \\ &= \mathbb{E}[X^p]\mathbb{E}[Y^q]^{p/q}.\end{aligned}$$

The original inequality $\|XY\|_1 \le \|X\|_p \|Y\|_q$, which is equivalent to $\mathbb{E}[XY] \le \mathbb{E}[X^p]^{1/p}\mathbb{E}[Y^q]^{1/q}$, follows by taking the $1/p$-th power of both sides of the above inequality.

For Young's inequality (c), by the condition $\frac{1}{p}+\frac{1}{q}=1$ and the weighted arithmetic-mean geometric-mean inequality, we have:

$$\frac{X^p}{p}+\frac{Y^q}{q} \ge (X^p)^{1/p}(Y^q)^{1/q} = XY.$$

Taking the expectation of both sides and applying the linearity of expectation, we obtain:

$$\begin{aligned}\mathbb{E}[XY] &\le \mathbb{E}\left[\frac{X^p}{p}+\frac{Y^q}{q}\right] \\ &= \frac{1}{p}\mathbb{E}\left[X^p\right]+\frac{1}{q}\mathbb{E}\left[Y^q\right],\end{aligned}$$

which is equivalent to $\|XY\|_1 \le \frac{1}{p}\|X\|_p^p + \frac{1}{q}\|Y\|_q^q$.

Problem 3.5 continued on next page...

For Minkowski's inequality (d), let q be the conjugate exponent of p, so that $\frac{1}{p} + \frac{1}{q} = 1$. We then have:

$$\begin{aligned}
\mathbb{E}[(X+Y)^p]^{1/p} &= \mathbb{E}[(X+Y)^p] \cdot \mathbb{E}[(X+Y)^p]^{-1/q} && (1 - 1/q = 1/p) \\
&= \mathbb{E}[(X+Y) \cdot (X+Y)^{p-1}] \cdot \mathbb{E}[(X+Y)^p]^{-1/q} \\
&= \big(\mathbb{E}[X \cdot (X+Y)^{p-1}] + \mathbb{E}[Y \cdot (X+Y)^{p-1}]\big) \cdot \mathbb{E}[(X+Y)^p]^{-1/q} && \text{(linearity of expectation)} \\
&\leq (\mathbb{E}[X^p]^{1/p} + \mathbb{E}[Y^p]^{1/p}) \cdot \mathbb{E}[(X+Y)^{(p-1)q}]^{1/q} \cdot \mathbb{E}[(X+Y)^p]^{-1/q} && \text{(Hölder's inequality)} \\
&= \mathbb{E}[X^p]^{1/p} + \mathbb{E}[Y^p]^{1/p}, && ((p-1)q = p)
\end{aligned}$$

which is equivalent to Minkowski's inequality $\|X+Y\|_p \leq \|X\|_p + \|Y\|_p$.

Problem 3.6

Let X be a random variable.

(a) Prove that for all $a \geq 0$,

$$\mathbb{E}\left[X \mathbb{1}_{[X>a]}\right] \leq \mathbb{E}\left[X^2\right]^{1/2} \Pr[X > a]^{1/2}.$$

(b) Suppose X is non-negative and $\mathbb{E}[X^2] \neq 0$. Prove that for all $\theta \in [0, 1]$,

$$\Pr[X > \theta \mathbb{E}[X]] \geq (1-\theta)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}\left[X^2\right]}.$$

(c) Suppose $\sigma(X) > 0$. Prove that for all $\alpha \in [0, 1]$,

$$\Pr[|X - \mathbb{E}[X]| > \alpha \cdot \sigma(X)] \geq \frac{\left(1-\alpha^2\right)^2}{\kappa(X)}.$$

where

$$\kappa(X) := \frac{\mathbb{E}\left[|X - \mathbb{E}[X]|^4\right]}{\sigma(X)^4}$$

is called the kurtosis of X.

Answer:

For part (a), we apply the Cauchy-Schwarz inequality (the special case of Hölder's inequality where $p = q = 2$, which was established in Problem 3.5) to obtain:

$$\begin{aligned}\mathbb{E}\left[X \mathbb{1}_{[X>a]}\right] &\leq \mathbb{E}\left[X^2\right]^{1/2} \mathbb{E}[\mathbb{1}_{[X>a]}^2]^{1/2} \\ &= \mathbb{E}\left[X^2\right]^{1/2} \Pr[X > a]^{1/2}.\end{aligned}$$

For part (b), we substitute $a = \theta \mathbb{E}[X]$ into the result from part (a) and rearrange terms to obtain:

$$\Pr[X > \theta \mathbb{E}[X]] \geq \frac{\mathbb{E}\left[X \mathbb{1}_{[X>\theta \mathbb{E}[X]]}\right]^2}{\mathbb{E}[X^2]}.$$

By the definition of expectation, we have $\mathbb{E}[X \mathbb{1}_{[X \leq a]}] = \mathbb{E}[X \mid X \leq a] \cdot \Pr[X \leq a] \leq a$. Therefore:

$$\begin{aligned}\mathbb{E}\left[X \mathbb{1}_{[X>\theta \mathbb{E}[X]]}\right] &= \mathbb{E}[X] - \mathbb{E}\left[X \mathbb{1}_{[X \leq \theta \mathbb{E}[X]]}\right] \\ &\geq \mathbb{E}[X] - \theta \mathbb{E}[X] \\ &= (1-\theta)\mathbb{E}[X].\end{aligned}$$

Combining these two inequalities, we obtain:

$$\Pr[X > \theta \mathbb{E}[X]] \geq (1-\theta)^2 \frac{\mathbb{E}\left[X\right]^2}{\mathbb{E}[X^2]}.$$

For part (c), let $f(X) = (X - \mathbb{E}[X])^2$. Note that $f(X)$ is non-negative, and $\mathbb{E}[f(X)] = \sigma(X)^2 \neq 0$ since

Problem 3.6 continued on next page...

$\sigma(X) > 0$. The left-hand side of the desired inequality can be rewritten as:

$$\Pr[|X - \mathbb{E}[X]| > \alpha \cdot \sigma(X)] = \Pr[f(X) > \alpha^2 \cdot \mathbb{E}[f(X)]].$$

Applying the result from part (b) to $f(X)$ with $\theta = \alpha^2$, we obtain:

$$\begin{aligned}
\Pr[f(X) > \alpha^2 \cdot \mathbb{E}[f(X)]] &\geq (1-\alpha^2)^2 \frac{\mathbb{E}[f(X)]^2}{\mathbb{E}[f(X)^2]} \\
&= (1-\alpha^2)^2 \frac{\sigma(X)^4}{\mathbb{E}[(X - \mathbb{E}[X])^4]} \\
&= \frac{(1-\alpha^2)^2}{\kappa(X)}.
\end{aligned}$$

Problem 3.7

Consider a random walk on a circle with vertices $0, 1, \ldots, n-1$. Initially, the drunkman is at vertex 0. At each step, a drunkman moves from the current vertex i to either $(i+1) \pmod n$ or $(i-1) \pmod n$ with equal probability. Prove that there exist universal constants $A > 0$ and $c > 0$ such that the probability that the drunkman has visited all vertices after t steps is at least $1 - Ae^{-ct/n^2}$ for all $t \geq 0$.

Answer:

We unfold the circle into the real line, and consider the corresponding simple symmetric random walk on the real line.

For any $t \geq 1$, let S_t denote the position of the walk after t steps, and let $X_i \in \{-1, 1\}$ denote the direction of the i-th step, so that $S_t = \sum_{i=1}^{t} X_i$.

If the walk moves by at least n steps along the line, it must cover all n vertices of the original circle. Therefore, the event that the drunkard visits all vertices on the circle is implied by the event that, for some time interval, the displacement of the walk on the unfolded real line has absolute value at least n.

By the property of variance, we have:

$$\operatorname{Var}(S_t) = t \operatorname{Var}(X_i) = t\left(\mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2\right) = t.$$

This implies $\sigma(S_t) = \sqrt{t}$ by definition. Since $\mathbb{E}[S_t] = t\mathbb{E}[X_i] = 0$, we obtain $\mathbb{E}[S_t^2] = \operatorname{Var}(S_t) + \mathbb{E}[S_t]^2 = t$. By the same method used for moment calculations in Problem 3.4, we can prove that

$$\mathbb{E}[S_t^4] = t\mathbb{E}[X_i^4] + 3t(t-1)\mathbb{E}[X_i^2]^2 = 3t^2 - 2t.$$

Substituting into the result from Problem 3.6(c), for all $\alpha \in [0, 1]$,

$$\Pr[|S_t| > \alpha \cdot \sigma(S_t)] \geq (1-\alpha^2)^2 \cdot \frac{\sigma(S_t)^4}{\mathbb{E}[S_t^4]},$$
$$\Pr\left[|S_t| > \alpha \cdot \sqrt{t}\right] \geq (1-\alpha^2)^2 \cdot \frac{t^2}{3t^2 - 2t} \geq \frac{(1-\alpha^2)^2}{3}.$$

We substitute $t = 2n^2$ and $\alpha^2 = \frac{1}{2}$ into the above inequality and obtain:

$$\Pr\left[|S_{2n^2}| > n\right] \geq \frac{(1-1/2)^2}{3} = \frac{1}{12}.$$

For any $t \geq 0$, we divide the t steps into $\left\lfloor \frac{t}{2n^2} \right\rfloor$ independent phases, each consisting of $2n^2$ steps. We ignore the remaining steps, which can only increase the probability of visiting all vertices. Let ΔS_k denote the displacement of the walk within the k-th phase. The distribution of ΔS_k is independent of the starting position of the phase, so $\Pr[|\Delta S_k| \geq n] \geq 1/12$ for all k.

Let T denote the event that the drunkard has visited all vertices after t steps. If any phase satisfies $|\Delta S_k| \geq n$, then T occurs. We thus have:

$$\Pr[T] = 1 - \Pr[\overline{T}] \geq 1 - \left(1 - \frac{1}{12}\right)^{\left\lfloor t/2n^2 \right\rfloor} \geq 1 - \left(\frac{11}{12}\right)^{t/(2n^2)-1} = 1 - Ae^{-ct/n^2},$$

where the universal constants are $A = \frac{12}{11}$ and $c = -\frac{1}{2}\ln\left(\frac{11}{12}\right)$. Note that $c > 0$ since $\ln(11/12) < 0$, satisfying the requirement that $A, c > 0$.

Problem 3.8

Prove Proposition 3.31. For convenience, you can only need to consider the case of $a \geq \mu$. That is, you only need to prove the following:

(a) $I(a) = +\infty$ if $a > u$;

(b) $0 \leq I(a) < +\infty$ for all $a \in [\mu, u]$;

(c) $I(\mu) = 0$.

(d) $I(a)$ is convex on the interval $[\mu, u]$: for any $a, b \in [\mu, u]$ and $\lambda \in [0, 1]$,

$$I(\lambda a + (1-\lambda)b) \leq \lambda I(a) + (1-\lambda) I(b).$$

Answer: For part (a), by definition, we have $M_X(t) = \mathbb{E}[e^{tX}] \leq e^{tu}$ for $t \geq 0$. For $t < 0$, $ta - \log M_X(t) \leq 0$, which does not contribute to the supremum. Therefore, for any $a > u$, we have:

$$\begin{aligned} I_X(a) &= \sup_{t \in \mathbb{R}} \{ta - \log M_X(t)\} \\ &\geq \sup_{t \geq 0} \{ta - tu\} = +\infty. \end{aligned}$$

For part (b), we first prove the lower bound. Substituting $t = 0$, we obtain:

$$I_X(a) \geq 0 \cdot a - \log M_X(0) = -\log 1 = 0.$$

For the upper bound, fix any $a \in [\mu, u]$. We may restrict our attention to $t \geq 0$, since for $t < 0$ and $a \geq \mu$, we have $ta - \log M_X(t) \leq 0$, which cannot exceed the value 0 attained at $t = 0$ and thus does not contribute to the supremum. Since e^{tX} is non-negative, we have $M_X(t) = \mathbb{E}[e^{tX}] \geq e^{tu} \cdot \Pr[X = u]$. Note that $\Pr[X = u] > 0$ since u is the maximum value that X can take. Therefore,

$$I_X(a) \leq \sup_{t \geq 0} \{tu - \log(e^{tu} \cdot \Pr[X = u])\} = -\ln \Pr[X = u] < +\infty.$$

For part (c), by Jensen's inequality applied to the convex function $f(x) = e^{tx}$, we have $\mathbb{E}[e^{tX}] \geq e^{t \cdot \mathbb{E}[X]} = e^{t\mu}$. Therefore,

$$I_X(\mu) = \sup_{t \in \mathbb{R}} \{t\mu - \log M_X(t)\} \leq \sup_{t \in \mathbb{R}} \{t\mu - t\mu\} = 0.$$

Since we have proved $I_X(\mu) \geq 0$ in part (b), we conclude $I_X(\mu) = 0$.

For part (d), by definition, we have:

$$\begin{aligned} I_X(\lambda a + (1-\lambda)b) &= \sup_{t \in \mathbb{R}} \{t[\lambda a + (1-\lambda)b] - \log M_X(t)\} \\ &= \sup_{t \in \mathbb{R}} \{\lambda[ta - \log M_X(t)] + (1-\lambda)[tb - \log M_X(t)]\} \\ &\leq \lambda \sup_{t \in \mathbb{R}} \{ta - \log M_X(t)\} + (1-\lambda) \sup_{t \in \mathbb{R}} \{tb - \log M_X(t)\} \\ &= \lambda I_X(a) + (1-\lambda) I_X(b). \end{aligned}$$

Problem 3.9

Let P be a math problem whose answer is either yes or no. Suppose you have trained an LLM that is good at problem solving. But P is very hard, so your LLM can only solve it with probability $1/2 + \epsilon$. To increase the probability of success, you can run the LLM multiple times and take the majority vote. That is, you run the LLM n times and answer yes if and only if at least half of the answers are yes. Prove that for some $n_{\min} = \mathcal{O}(\frac{1}{\epsilon^2} \log \frac{1}{\delta})$ as $(\epsilon, \delta) \to (0, 0)$, if you run the LLM $n \geq n_{\min}$ times, then the majority vote is correct with probability at least $1 - \delta$.

Answer:

Let $X_i \in \{-1, 1\}$ denote the result of the i-th run, where 1 represents a correct answer, while -1 represents an incorrect answer. The majority vote is correct if and only if $S_n = \sum_{i=1}^{n} X_i \geq 0$. By definition, we have $\mathbb{E}[X_i] = (1/2 + \epsilon) - (1/2 - \epsilon) = 2\epsilon$ and $\mathbb{E}[S_n] = n\mathbb{E}[X_i] = 2n\epsilon$.

By Hoeffding's inequality for independent bounded random variables, we have:

$$\Pr[S_n < 0] \leq \Pr\left[S_n \leq \mathbb{E}[S_n] - 2n\epsilon\right] \leq e^{-2\cdot(2n\epsilon)^2/(n\cdot 2^2)} = e^{-2n\epsilon^2}.$$

Therefore, for $n_{\min} = \frac{1}{2\epsilon^2} \log \frac{1}{\delta} = \mathcal{O}\left(\frac{1}{\epsilon^2} \log \frac{1}{\delta}\right)$ and any $n \geq n_{\min}$, we have:

$$\Pr[S_n \geq 0] = 1 - \Pr[S_n < 0] \geq 1 - e^{-2n\epsilon^2} \geq 1 - e^{-2n_{\min}\epsilon^2} = 1 - \delta.$$

Problem 3.10

Let G be a random graph drawn from the Erdős–Rényi model $G(n, p)$. That is, G has n vertices, and for each pair of distinct vertices $i \neq j$, between them an undirected edge is included with probability p. A coloring of G is an assignment of each vertex with a color such that no two adjacent vertices have the same color. The chromatic number $\chi(G)$ is the minimum number of colors needed to color the graph G. Show that for all $\alpha \geq 0$,

$$\Pr[|\chi(G) - \mathbb{E}[\chi(G)]| \geq \alpha\sqrt{n}] \leq 2e^{-2\alpha^2}.$$

Answer:

For every integer $2 \leq i \leq n$, let $X_i = (E_{i,1}, E_{i,2}, \ldots, E_{i,i-1})$ denote the vector of indicator variables for the presence of edges between vertex i and vertices $1, 2, \ldots, i-1$. Define $f(X_2, X_3, \ldots, X_n) := \chi(G)$, where G is the graph described by $X_2, X_3, \ldots, X_n$.

For each $2 \leq i \leq n$, we have the bounded difference property:

$$|f(x_2, \ldots, x_i, \ldots, x_n) - f(x_2, \ldots, x_i', \ldots, x_n)| \leq 1$$

for any fixed values of $x_2, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n$ and any possible values x_i, x_i'. This holds because modifying the adjacency of vertex i can change the chromatic number by at most 1: we can always color vertex i with a new color to obtain a valid coloring of the modified graph, so the chromatic number can increase by at most 1, and similarly can decrease by at most 1.

Since $X_2, X_3, \ldots, X_n$ are independent random variables (corresponding to disjoint edge sets), applying McDiarmid's inequality, we immediately obtain:

$$\Pr[|\chi(G) - \mathbb{E}[\chi(G)]| \geq \alpha\sqrt{n}] \leq 2e^{-2(\alpha\sqrt{n})^2/L^2} \leq 2e^{-2\alpha^2},$$

where $L^2 = \sum_{i=2}^{n} 1^2 = n - 1$.

Problem 3.11

Following McDiarmid's inequality, let $X_1, \ldots, X_n$ be independent random variables, and let $f : \mathbb{R}^n \to \mathbb{R}$ be a function. Assume that for each $i = 1, \ldots, n$ there is a constant $c_i \geq 0$ such that whenever two inputs differ only in coordinate i,

$$|f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_n)| \leq c_i.$$

Let $L^2 := \sum_{i=1}^n c_i^2$. Now, instead of giving a high-probability tail bound, prove the following upper bound on the variance of $f(X_1, \ldots, X_n)$:

$$\mathrm{Var}\left[f(X_1, \ldots, X_n)\right] \leq \frac{1}{4} L^2.$$

Answer:

Let $F := f(X_1, \ldots, X_n)$, and define the Doob martingale

$$Z_i := \mathbb{E}[F \mid X_1, \ldots, X_i], \quad Z_0 := \mathbb{E}[F].$$

Let $\Delta_i = Z_i - Z_{i-1}$ denote the martingale increments, then $Z_n - Z_0 = \sum_{i=1}^n \Delta_i$ and $F = Z_n$. By the definition of variance and the linearity of expectation, we have:

$$\begin{aligned}
\mathrm{Var}(F) &= \mathbb{E}\left[(Z_n - Z_0)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{i=1}^n \Delta_i\right)^2\right] \\
&= \sum_{i=1}^n \mathbb{E}\left[\Delta_i^2\right] + 2 \sum_{1 \leq i < j \leq n} \mathbb{E}[\Delta_i \Delta_j].
\end{aligned}$$

Consider the cross terms. For any $i < j$, by the law of total expectation, we have:

$$\begin{aligned}
\mathbb{E}[\Delta_i \Delta_j] &= \mathbb{E}\left[\mathbb{E}\left[\Delta_i \Delta_j \mid X_{<j}\right]\right] \\
&= \mathbb{E}\left[\Delta_i \cdot \mathbb{E}\left[\Delta_j \mid X_{<j}\right]\right].
\end{aligned}$$

Since Z is a martingale, we have $\mathbb{E}[\Delta_j \mid X_{<j}] = 0$, which implies $\mathbb{E}[\Delta_i \Delta_j] = 0$.

We now compute the upper bound for $\mathbb{E}\left[\Delta_i^2\right]$. From the lecture notes, we derive that Δ_i lies in an interval of length at most c_i. Let t_i be the midpoint of the interval. Consider the expectation $\mathbb{E}\left[(\Delta_i - t_i)^2\right]$:

$$\mathbb{E}\left[(\Delta_i - t_i)^2\right] = \mathbb{E}\left[\Delta_i^2\right] - 2t_i \mathbb{E}[\Delta_i] + t_i^2 \geq \mathbb{E}\left[\Delta_i^2\right],$$

where $\mathbb{E}[\Delta_i] = \mathbb{E}\left[\mathbb{E}[\Delta_i \mid X_{<i}]\right] = 0$. Since t_i is the midpoint of the interval, we have $|\Delta_i - t_i| \leq \frac{c_i}{2}$, hence $\mathbb{E}\left[\Delta_i^2\right] \leq \mathbb{E}\left[(\Delta_i - t_i)^2\right] \leq \frac{c_i^2}{4}$. Substituting this bound into the expression for $\mathrm{Var}(F)$, we finally obtain:

$$\mathrm{Var}\left[f(X_1, \ldots, X_n)\right] \leq \sum_{i=1}^n \frac{c_i^2}{4} = \frac{L^2}{4}.$$